SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 5

Total Marks:100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Sri Mathi. B (192511448) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Srimathi. B

COI Short Answer	Name : B. SriMathi Reg.No : 192511448 Course code : CSA0414 Course Name : Operating System

Modern operating System - Multiple users and Processes

a. Types of Operating systems

For a modern system supporting multiple users, processors, CPU, memory and I/O devices, the suitable OS types are.

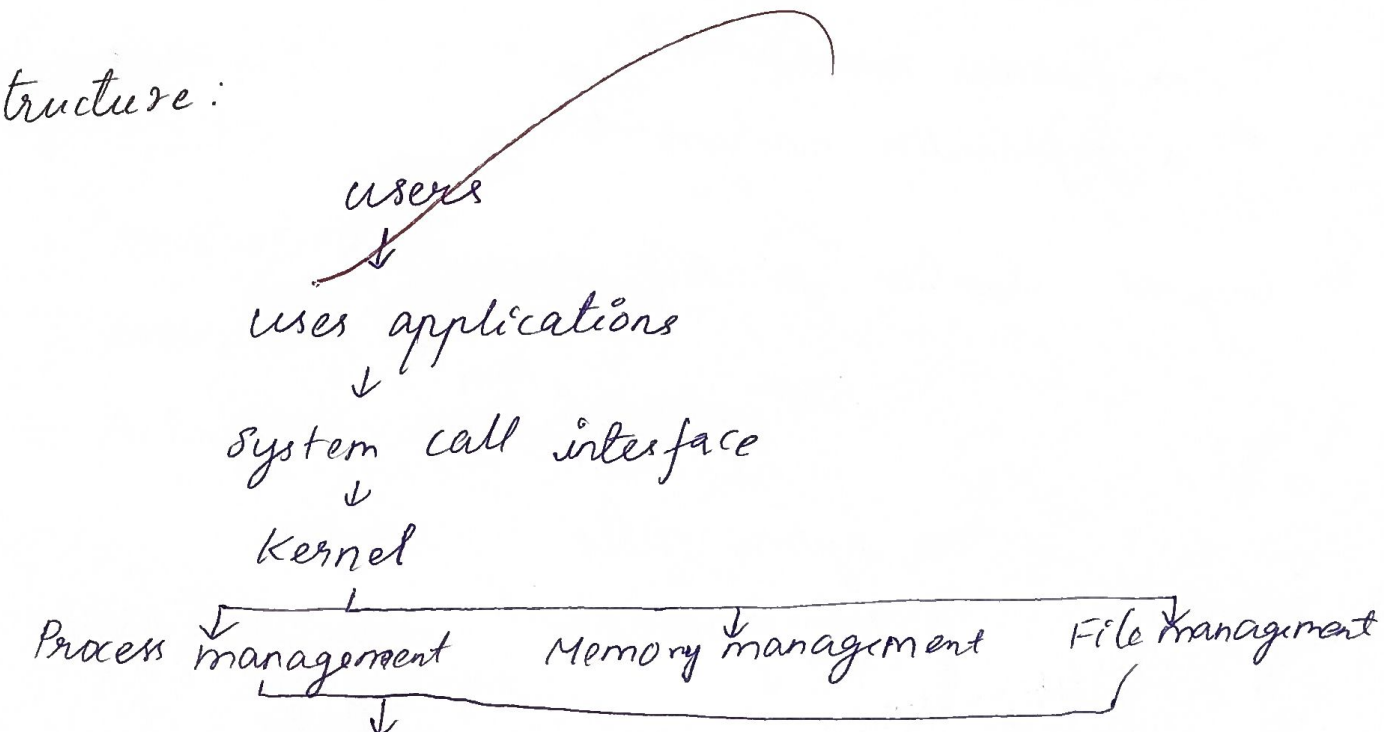
1. Multi-user operating system - allows multiple users to access system resources simultaneously.

2. Multiprogramming / Multitasking OS - allows multiple processes to execute concurrently by sharing CPU time.

3. Time-sharing OS - gives each user / process a small CPU time slice for interactive operation.

4. Network OS manages should shared resources such as files

b. OS structure:



The kernel manages CPU scheduling, memory, files and I/O devices.

c) System calls to create and execute a Process

In Unix / Linux:

```
pid = fork();
if (pid == 0)
{
  execl ("/bin/program", "program",
NULL);
}
else
{
  wait (NULL);
}
```

Sequence: fork() -> exec() -> wait()

---

2. File & Device Management

a. Categories of system calls.

=> File management
=> Device management
=> Process management
=> Information management

b. Layered structure for File operations

user application
↓
system calls
open / read / write
↓
File system

I/O Management

↓

Device Drivers

↓

Hardware

Open, Read and close a file

```
int fd;
char buffer[100];
fd = open("data.text", O_RDONLY);
read(fd, buffer, 100);
close(fd);
```

sequence : open() → read() → close()

Real-Time Operating System

a2. Type of OS & Suitability

-> The suitable OS is a Real-Time operating system (RTOS).

-> An RTOS is designed to respond to events within a specified time limit. It is suitable for industrial machinery.

example: => Industrial control systems

=> Robotics

=> Medical equipment

=> Embedded systems

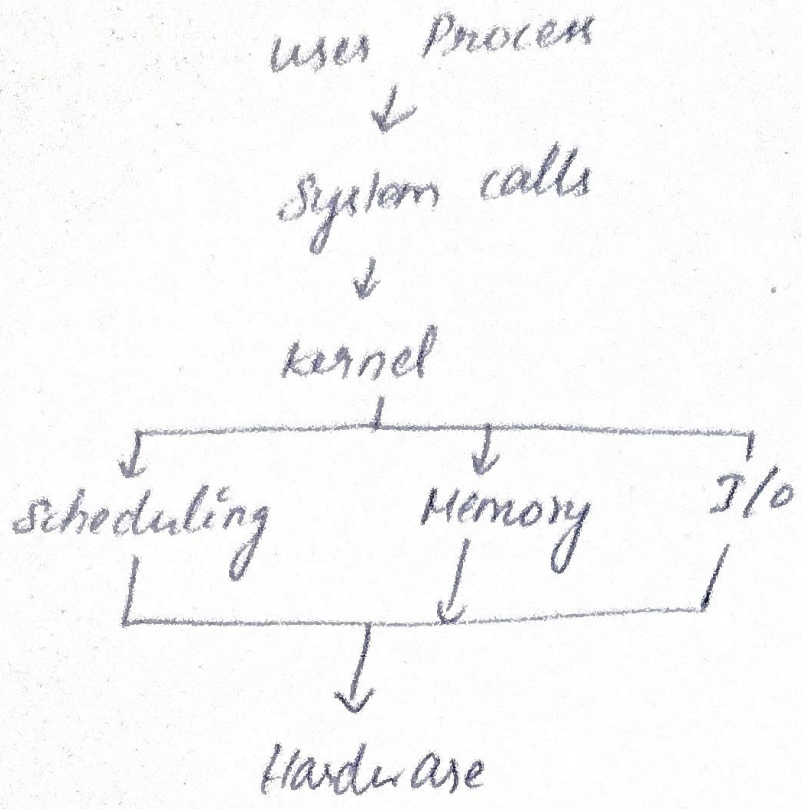
=> Automobile control system

and syst
BL2, BL3)

Total Marks:100

ssion is my original work
nt.
t in disciplinary action.

b. Hardware kernel . Process Interaction.



c. System calls for Process Synchronization.

1. wait () - allows a proces to wait for another process
2. Signal () - signals another process / thread.
3. Semaphone operations - control access to shared resources
4. Pthread_muten_lock() - locks a shared resources
5. pthread - muter - unlak() - releases the resousce

eg: pthread - muten - lock (& muten):

---

4. Distributed Operating System:

a. Classification & comparison.

Distributed as manages multiple connected computers and shares their resources.

a. Distributed: Multiple computer, resource sharing
Centralized: One main computer, centralized resource

b.

Node 1 ⟷ Network ⟷ Node 2

1

Node 3

c. IPC sequence:

Send() → communication channel → recv() → close()

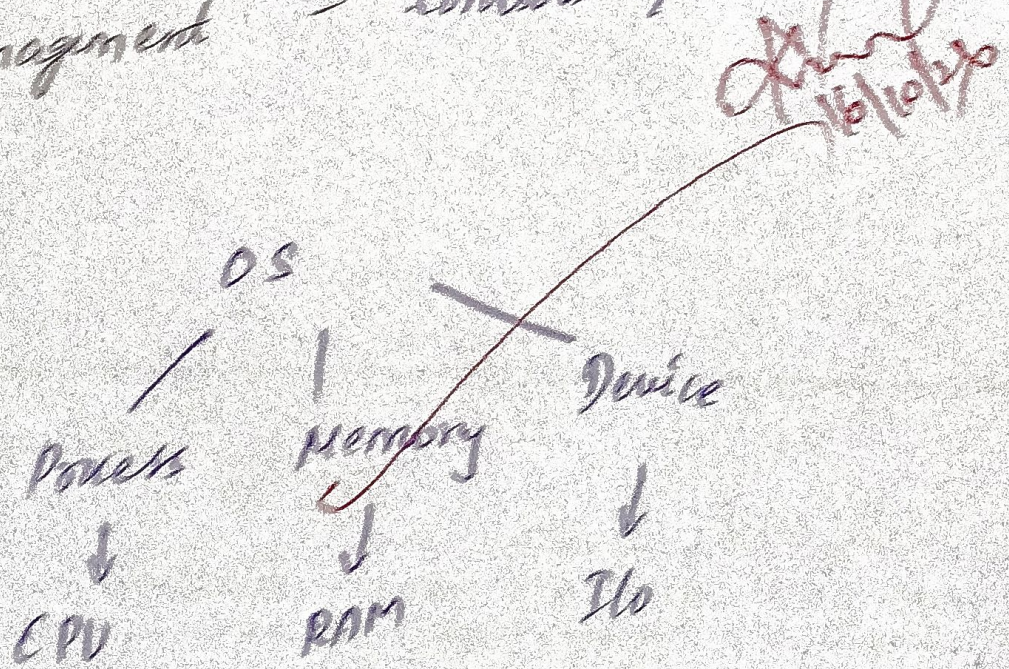
---

2. OS management

a. Major functions:

- Process management – CPU scheduling
- Memory management – RAM allocation
- Device management – controls I/O devices

b.



c.

Memory: mmap(), munmap()

Device: open(), read(), write(), ioctl(), close()